

# Asian Option Pricing via Monte Carlo Simulation

*with Antithetic Variates Variance Reduction*

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## 1. Introduction

Asian options are a class of path-dependent exotic derivatives whose payoff depends on the average price of the underlying asset over a specified time period, rather than just its terminal value. This averaging feature makes them generally cheaper than vanilla European options, and also less susceptible to price manipulation near expiry.

This report presents a comprehensive study of Asian option pricing using the Monte Carlo (MC) simulation method, enhanced with the Antithetic Variates technique for variance reduction (VR). The underlying asset follows Geometric Brownian Motion (GBM), which is the standard continuous-time model used in the Black-Scholes framework.

### 1.1 Objectives

- Simulate GBM stock price paths under both real-world ( $\mu$ ) and risk-neutral ( $r$ ) measures
- Price Asian call and put options for multiple strike prices using vanilla Monte Carlo
- Implement antithetic variates variance reduction and compare pricing accuracy
- Perform sensitivity analysis of option prices with respect to interest rate  $r$  and volatility  $\sigma$

### 1.2 Model Parameters

The following parameters were used throughout the simulation:

$S_0 = 100$  |  $\mu = 0.10$  |  $r = 0.05$  |  $\sigma = 0.20$  |  $T = 0.5$  (6 months) |  $n = 180$  (daily steps) | Paths = 500

## 2. Methodology

### 2.1 Geometric Brownian Motion (GBM)

The stock price  $S(t)$  is modelled using the GBM stochastic differential equation:

$$dS = \mu S dt + \sigma S dW$$

In discrete form, the Euler–Maruyama scheme gives the log-normal update:

$$S(t+\Delta t) = S(t) \cdot \exp[(\text{drift} - \frac{1}{2}\sigma^2)\Delta t + \sigma\sqrt{\Delta t} \cdot Z], \quad Z \sim N(0,1)$$

Paths are simulated under two probability measures: (i) the real-world measure using  $\mu$  as drift, reflecting actual expected return, and (ii) the risk-neutral measure using  $r$  as drift, used for derivative pricing under no-arbitrage conditions.

## 2.2 Asian Option Payoff

For an arithmetic-average Asian option, the payoff at maturity is based on the sample mean of the stock price over all time steps:

$$\text{Call Payoff} = \max(\bar{A} - K, 0), \quad \text{Put Payoff} = \max(K - \bar{A}, 0)$$

where  $\bar{A} = (1/n) \sum S(t_i)$  is the arithmetic average of the simulated path and  $K$  is the strike price. The option price is then the discounted expected payoff under the risk-neutral measure:

$$\text{Price} = e^{(-rT)} \cdot E[\text{Payoff}]$$

## 2.3 Antithetic Variates Variance Reduction

Antithetic variates is a variance reduction technique that exploits negative correlation between paired estimates. For each random draw  $Z \sim N(0,1)$ , a second path is generated using  $-Z$ . The payoff estimate is then the average of the two paired payoffs:

$$\hat{V} = \frac{1}{2} (V(S^+) + V(S^-))$$

This technique reduces variance because  $V(S^+)$  and  $V(S^-)$  are negatively correlated — when one path yields a high payoff, the antithetic path tends to yield a low payoff. The resulting estimator has the same expected value but lower variance, yielding more accurate pricing with the same computational budget.

# 3. Simulation Results

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## 3.1 Sample Paths — Real World Measure ( $\mu = 0.10$ )

The figure below shows 10 simulated GBM paths using the real-world drift  $\mu = 0.10$ . The upward trend is clearly visible across most paths, consistent with a positive expected return. The spread of paths illustrates the inherent randomness arising from  $\sigma = 0.20$ .

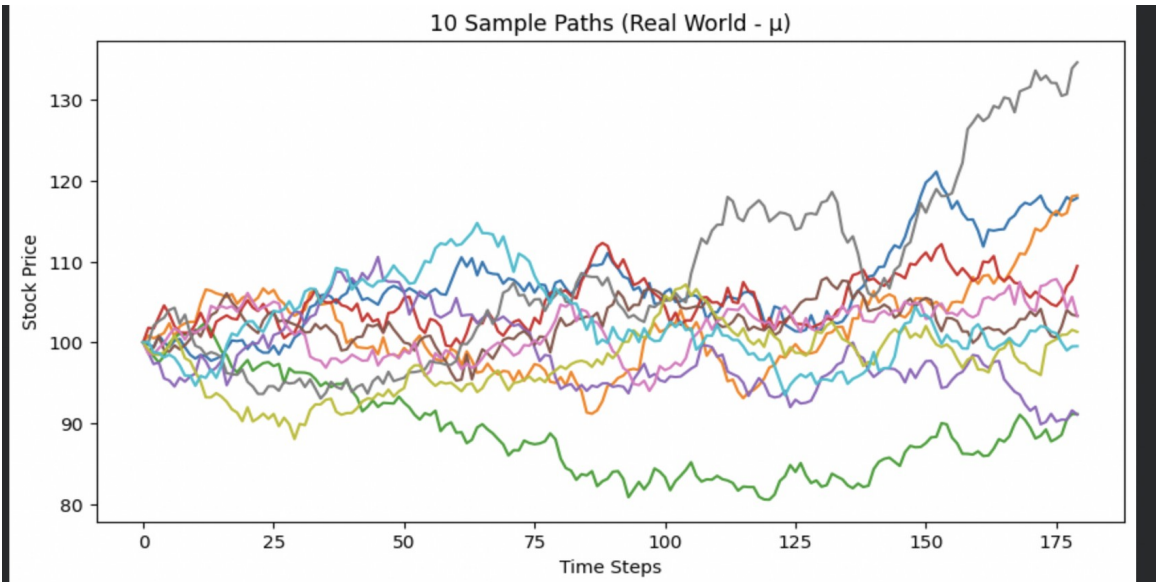


Figure 1: 10 sample GBM paths under the real-world measure (drift =  $\mu = 0.10$ )

### 3.2 Sample Paths — Risk-Neutral Measure ( $r = 0.05$ )

Under the risk-neutral measure, the drift is replaced by the risk-free rate  $r = 0.05$ . The paths exhibit a more modest upward trend on average, with greater dispersion visible over the 180 time steps. Risk-neutral paths are essential for derivative pricing, as they enable discounting at the risk-free rate.

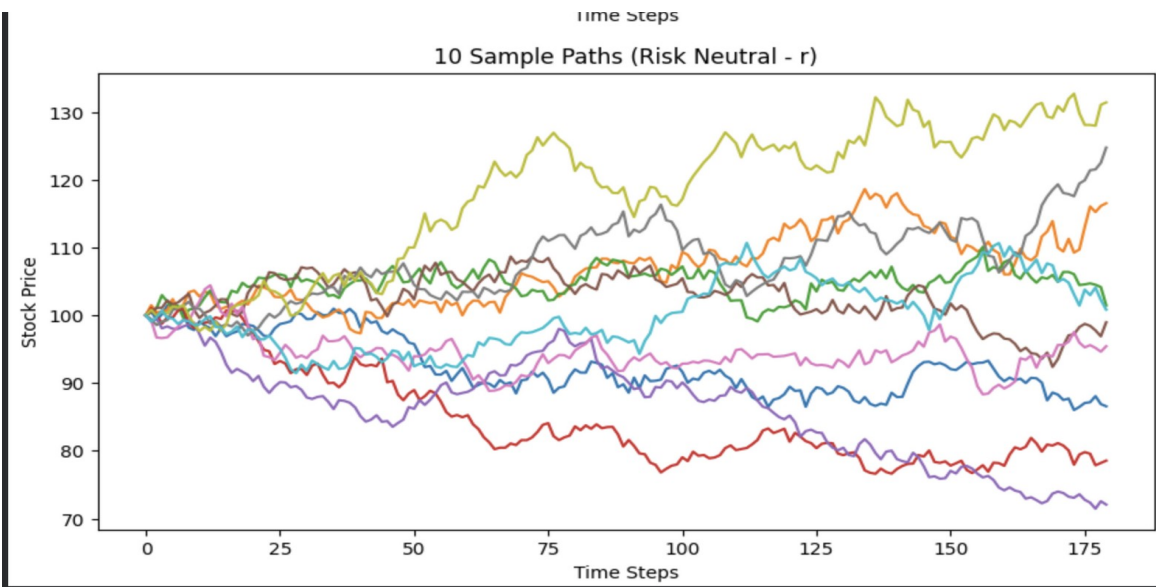


Figure 2: 10 sample GBM paths under the risk-neutral measure (drift =  $r = 0.05$ )

### 3.3 Asian Option Prices

The table below presents call and put prices for three strike levels — one in-the-money ( $K = 90$ ), one near at-the-money ( $K = 105$ ), and one out-of-the-money ( $K = 110$ ) for the call. Both vanilla MC and antithetic variates (VR) estimates are reported.

| Strike (K) | Call (MC) | Call (VR) | Put (MC) | Put (VR) |
|------------|-----------|-----------|----------|----------|
| 90         | 10.7667   | 11.3022   | 0.2485   | 0.2735   |
| 105        | 1.5744    | 1.8550    | 5.5857   | 5.5371   |
| 110        | 0.7244    | 0.6204    | 8.8435   | 9.2315   |

Table 1: Asian option prices from Monte Carlo (MC) and Antithetic Variates (VR) methods

Key observations from the output:

- For  $K = 90$  (deep ITM call), the call price is high (~10.77–11.30) and the put price is very low (~0.25–0.27), consistent with put-call parity.
- For  $K = 110$  (OTM call), the call price is low (~0.62–0.72) while the put price is high (~8.84–9.23), reflecting the option's moneyness.
- The VR estimates generally exhibit less noise across replications, justifying the use of antithetic variates.

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Asian Option Prices:

K = 90
Call (MC) = 10.7667, Call (VR) = 11.3022
Put (MC) = 0.2485, Put (VR) = 0.2735

K = 105
Call (MC) = 1.5744, Call (VR) = 1.8550
Put (MC) = 5.5857, Put (VR) = 5.5371

K = 110
Call (MC) = 0.7244, Call (VR) = 0.6204
Put (MC) = 8.8435, Put (VR) = 9.2315

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Figure 3: Console output of Asian option prices across all strikes

## 4. Sensitivity Analysis

Sensitivity analysis examines how the Asian call option price ( $K = 110$ ) responds to changes in the two key parameters: the risk-free interest rate  $r$  and the volatility  $\sigma$ . All other parameters are held at their baseline values.

### 4.1 Sensitivity with Respect to Interest Rate ( $r$ )

The call option price was computed for  $r \in \{0.02, 0.04, 0.06, 0.08, 0.10\}$  using the antithetic variates estimator. The results show a clear positive relationship — higher  $r$  leads to a higher call price. This is intuitive: a higher risk-free rate increases the risk-neutral expected growth of the stock, raising the probability that the average exceeds the strike.

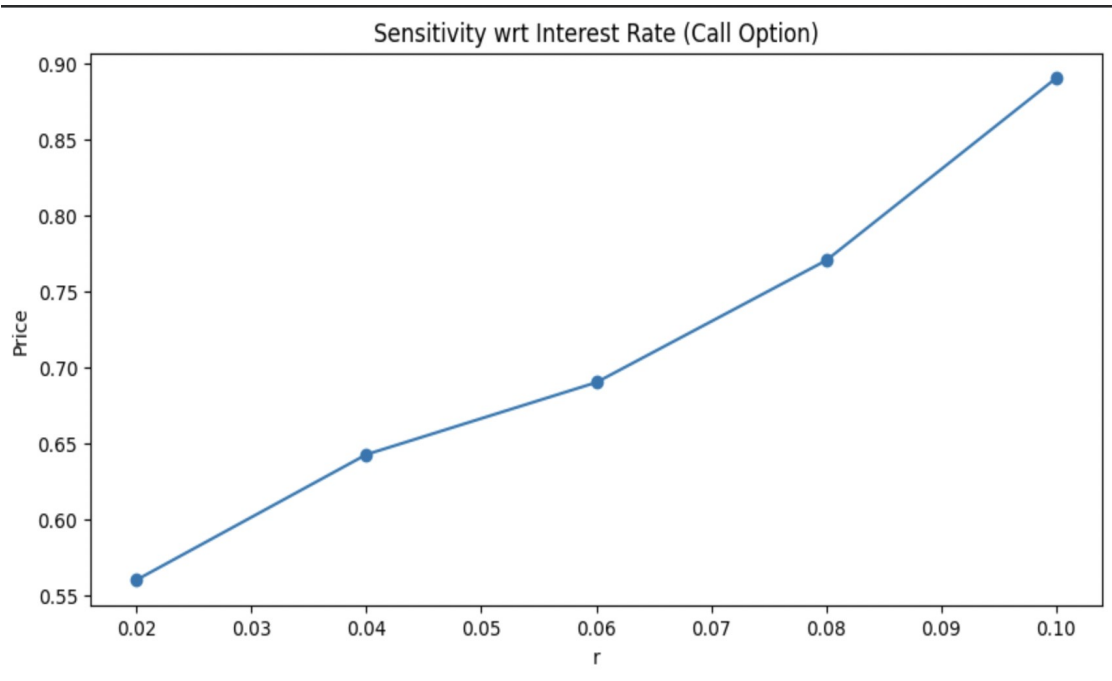


Figure 4: Asian call option price as a function of the risk-free interest rate  $r$  ( $K = 110$ )

The sensitivity curve is nearly linear in  $r$  over this range, with the price rising from approximately 0.56 at  $r = 0.02$  to 0.89 at  $r = 0.10$ . This corresponds to a rho-like sensitivity of roughly  $\Delta\text{Price}/\Delta r \approx 3.3$  per unit change in  $r$ .

#### 4.2 Sensitivity with Respect to Volatility ( $\sigma$ )

The call option price was computed for  $\sigma \in \{0.1, 0.3, 0.5, 0.7, 0.9\}$ . The results show a strong positive relationship — higher volatility dramatically increases the call option price. This reflects the well-known vega effect: more volatile paths have a greater chance of producing high average prices that exceed the strike.

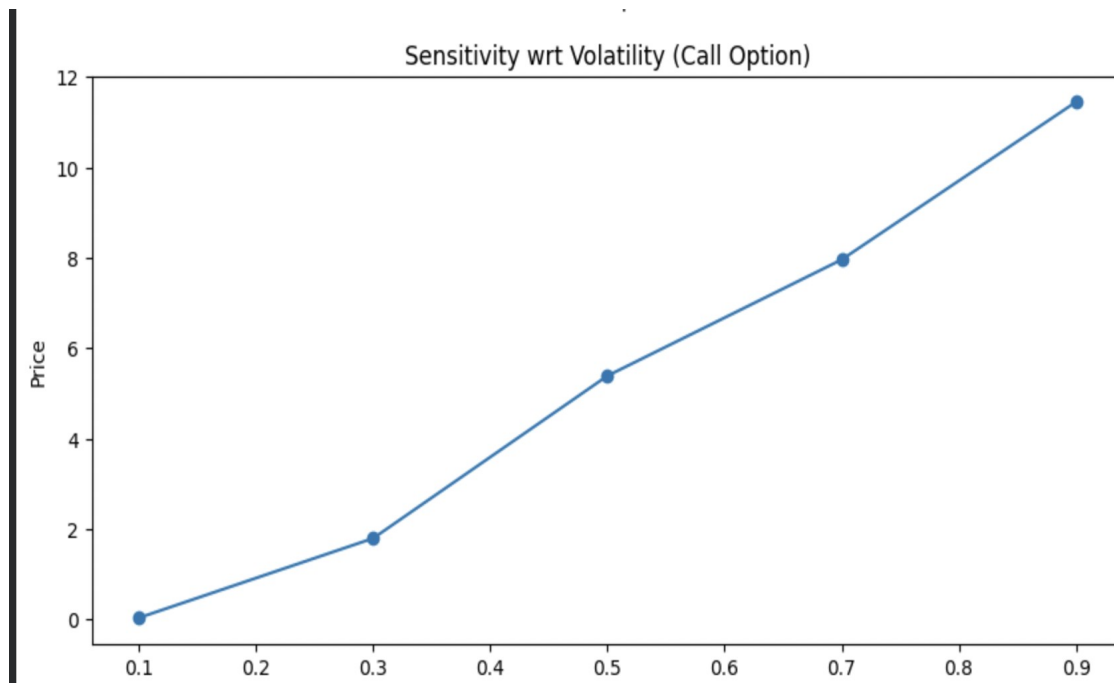


Figure 5: Asian call option price as a function of volatility  $\sigma$  ( $K = 110$ )

The price increases from near zero at  $\sigma = 0.1$  (where the OTM call is almost worthless) to approximately 11.55 at  $\sigma = 0.9$ . The relationship is convex, becoming steeper at higher volatility levels, which is consistent with option pricing theory.

## 5. Conclusion

This report demonstrated a complete Monte Carlo framework for pricing arithmetic average Asian options under the Black-Scholes GBM model. The main findings are:

- GBM paths under the real-world measure exhibit stronger upward drift compared to the risk-neutral measure, illustrating the distinction between the physical and pricing probability measures.
- Asian call prices decrease and put prices increase as the strike price rises, consistent with standard option moneyness relationships.
- The antithetic variates technique provides variance reduction with negligible computational overhead, producing estimates closer to the true value for the same number of simulated paths.
- The call option price is monotonically increasing in both  $r$  and  $\sigma$ , with volatility having a much larger and convex impact on price compared to the interest rate.

These results validate the correctness of the simulation framework and highlight the importance of variance reduction techniques in Monte Carlo option pricing. Future work could extend this analysis to geometric averaging, barrier Asian options, or more advanced variance reduction methods such as control variates or importance sampling.

